HEFTNet: modelling the di-Higgs m_{hh} spectrum and its uncertainty

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Executive summary and roadmap, October 31, 2026

Goal. A fast surrogate for POWHEG: the cross section per 15 GeV bin of m_{hh} (255–1455 GeV) as a function of five HEFT coefficients, with a calibrated uncertainty. `HEFTNet` writes $\sigma = \sum_{i=1}^{23} C_i(\theta)\, a_i(m_{hh})$, with known polynomials C_i and 23 functions a_i given by a network of 3,944 parameters that sees only m_{hh}. Training: 3,989 POWHEG spectra (274,152 rows used); validation on 1,000 independent test spectra.

Principal findings

1. **The data are exact Poisson counts.** Every spectrum is $q_s\times$ integer counts (40,000 generated events per run; a few incomplete runs, listed in `data/README.md`). The noise is known exactly; the earlier estimate (N_{eff}) was within 3% for typical spectra.
2. **Weight the errors by the noise.** L1 training leaves the threshold region poor ($\langle\chi^2\rangle = 8.4$ per bin at 255–555 GeV). A fine-tune with the Huber loss of the pulls ($\delta = 2$) brings it to 1.6 (insensitive to δ), with median $\chi^2/N = 1.07$ on the test spectra.
3. **The ensemble spread is mostly optimisation, and it under-covers.** 20-network ensembles (seeds; bootstrap of rows): 80% of the fine-tuned spread is due to the random seed, 20% to the finite sample; the spread is 5.6% of the POWHEG noise, and the pulls have s.d. 1.07 (1.15 near threshold).
4. **The model error dominates.** A discrepancy term $a(m_{hh})\sigma_k^2$, with $a = 0.76\, e^{-(m_{hh}-255)/76\,\text{GeV}} + 0.11$ fitted on held-out training rows, i.e. 0.3–0.9 of the noise, gives pull s.d. 1.003 (1.02 near threshold) and 95.1% of bins with $|z| < 2$ (95.4% expected).
5. **Bayes works, once computed correctly.** HMC (Neal's method) gave a good mean but a width of 0.5% of the noise, because a single chain could not explore a posterior with curvature ratios $\sim 10^6$. A mixture of linearised Laplace approximations about the 20 networks (exact for this architecture, 30 s per network on a CPU) gives 2.4% (within networks) $\oplus$ 5.6% (between) = 6.1% of the noise; the prior hardly matters ($\pm$4%). The posterior is narrow because $\sim$400 effective parameters are fixed by 2.8×10^5 rows; calibration requires the model-error term.
6. **Where the surrogate fails**: the worst test spectra are those with the smallest total cross section (strong destructive interference), e.g. $\chi^2/N = 68$ at the smallest-σ test point.

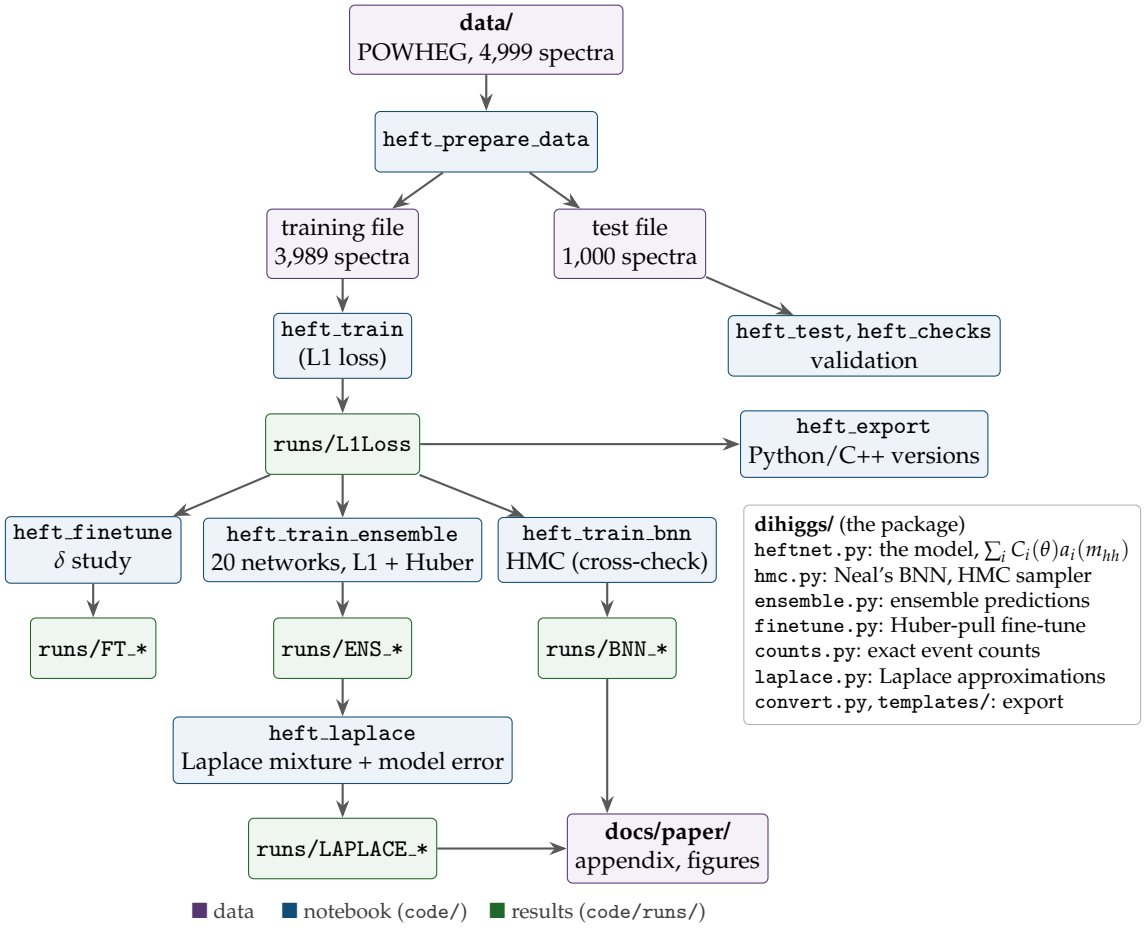
Investigations, in order.

- *BNN with HMC* (`hmc.py`): Laplace and Gaussian likelihoods, 10% and full data, longer trajectories; diagnosis in `docs/bnn/bnn_plan2.pdf`.
- *Ensembles* (`ensemble.py`): seeds vs bootstrap, law-of-total-variance decomposition, the reference-ensemble question (`docs/ensemble`).
- *Huber-pull fine-tune* (`finetune.py`): $\delta = 1, 2, 3, \infty$; 10,000 full-batch steps (`docs/huber`).
- *Exact counts* (`counts.py`): scale factor and integer counts per spectrum; empty bins restored (`docs/counts`).
- *Mixture of Laplace approximations* (`laplace.py`): MacKay's evidence for the prior, linearisation checks (`docs/bnn/laplace_explained.pdf`).

Recommended result for the paper. The fine-tuned ensemble mean, with uncertainty = ensemble spread (or the Laplace mixture) $\oplus$ model error; the Bayesian mixture as a consistent cross-check. Draft appendix and the 9-panel spectra figure: `docs/paper`.

Open items. Re-quote the appendix with the exact Poisson noise (shifts of $\sim$1%); the model error is treated as independent between bins (a correlated, Gaussian-process discrepancy is the principled extension); the short POWHEG runs listed in `data/README.md` are kept in the default training file; consider more training points where the cross section is small.

The `dihiggs` repository at a glance



Roadmap of the documents (all in `docs/`; read top to bottom, or jump to the topic you need)

Start here	
`paper/executive_summary.pdf`	this summary
Physics and the model	
`theory.pdf`	LHC Higgs WG note on EFT descriptions of hh production (arXiv:2304.01968): the C_i
`dihiggs_model.pdf`	the `HEFTNet` model: exact θ dependence, a one-dimensional learning problem
`spectra.pdf`	one-page overview of the POWHEG spectra
Data, pipeline and checks	
`report/dihiggs_report.pdf`	the pipeline from POWHEG files to trained and tested model
`checks_plan.pdf`	three checks of the data and test pipeline
`counts/counts_note.pdf`	the spectra are scaled Poisson counts: exact noise
Training	
`huber/huber_plan.pdf`	the Huber-pull fine-tune (the accuracy gain near threshold)
`curriculum_plan.pdf, _v2.pdf`	side study: curriculum learning
Uncertainty	
`ensemble/ensemble_plan.pdf`	seeds and bootstrap ensembles; the reference-ensemble question
`bnn/bnn_plan.pdf`	design of the Bayesian network and the HMC module
`bnn/bnn_run1.pdf–run3.pdf`	the three HMC runs (incl. a primer on mixing)
`bnn/bnn_plan2.pdf`	why HMC gave too-small widths; the Laplace plan
`bnn/laplace_explained.pdf`	the mixture of Laplace approximations, explained, with results
Paper	
`paper/appendix_draft.pdf`	draft JHEP appendix on the uncertainty
`paper/fig_*.pdf`	paper figures: 9 spectra (`fig_spectra_panel`); the 23 $a_i(m_{hh})$ (`fig_coefficients`)
`paper/all_test_spectra.pdf`	all 1,000 test spectra against the model (51 pages)
Practical	
`remote_gpu.pdf`	running the notebooks on the RTX 5090 (`tmux`, `nohup`)